

Bitcoin Market Sentiment and Trader Performance Analysis

Submitted By

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1. Introduction

Financial markets are heavily influenced by investor emotions and sentiment. The Bitcoin Fear & Greed Index is a popular indicator used to measure market psychology, ranging from Extreme Fear to Extreme Greed. Understanding how market sentiment affects trader performance can help identify profitable trading opportunities and improve risk management strategies.

This project analyzes the relationship between Bitcoin market sentiment and trader performance using two datasets:

1. Bitcoin Fear & Greed Index Dataset
2. Historical Hyperliquid Trader Dataset

The objective is to discover patterns between market sentiment and trading outcomes and provide data-driven insights for smarter trading decisions.

2. Dataset Description

Dataset 1: Bitcoin Fear & Greed Index

Features:

- Timestamp
- Value
- Classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed)
- Date

This dataset provides daily sentiment classifications representing the emotional state of the cryptocurrency market.

Dataset 2: Hyperliquid Historical Trader Data

Features:

- Account
- Coin
- Execution Price
- Size Tokens
- Size USD
- Side (Buy/Sell)
- Timestamp IST
- Start Position
- Direction
- Closed PnL
- Fee
- Transaction Information

This dataset contains historical trading activity and profitability information.

3. Methodology

The following steps were performed:

Data Cleaning

- Converted timestamps into datetime format.
- Extracted trading dates from timestamps.
- Standardized column names.

Data Integration

- Merged the Fear & Greed dataset with trading data using the Date column.
- Assigned a market sentiment classification to each trade.

Feature Engineering

- Created a Win column where:
 - Win = True if Closed PnL > 0
 - Win = False otherwise

Exploratory Data Analysis (EDA)

- Average Profit by Sentiment
- Total Profit by Sentiment
- Trading Activity Analysis

- Win Rate Analysis
- Trade Size Analysis
- PnL Distribution Analysis

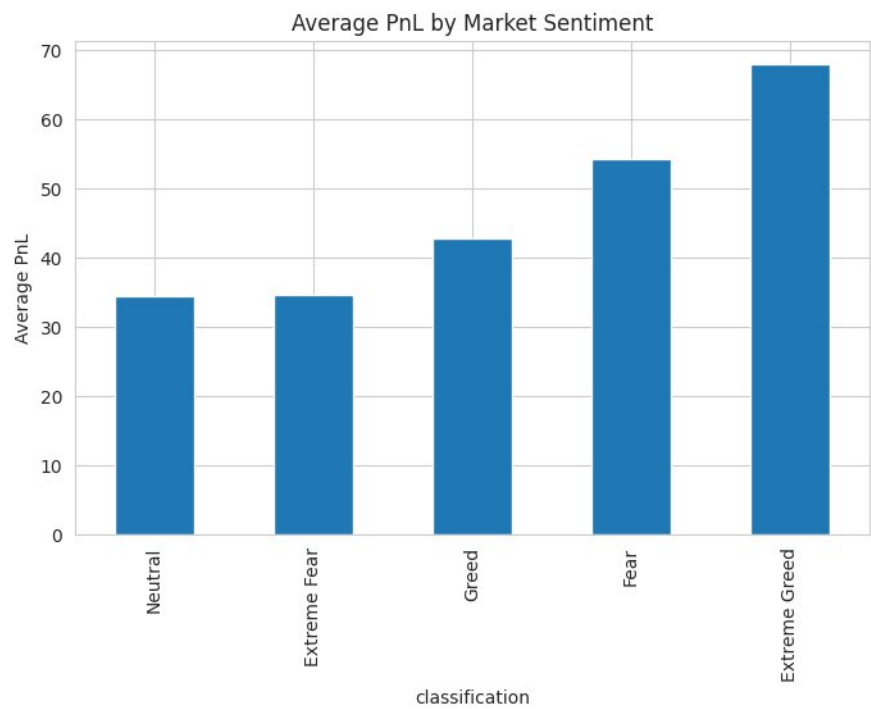
4. Results and Analysis

Summary Statistics

Sentiment	Trade Count	Average PnL	Total PnL
Extreme Fear	21,400	34.54	739,110
Fear	61,837	54.29	3,357,155
Neutral	37,686	34.31	1,292,921
Greed	50,303	42.74	2,150,129
Extreme Greed	39,992	67.89	2,715,171

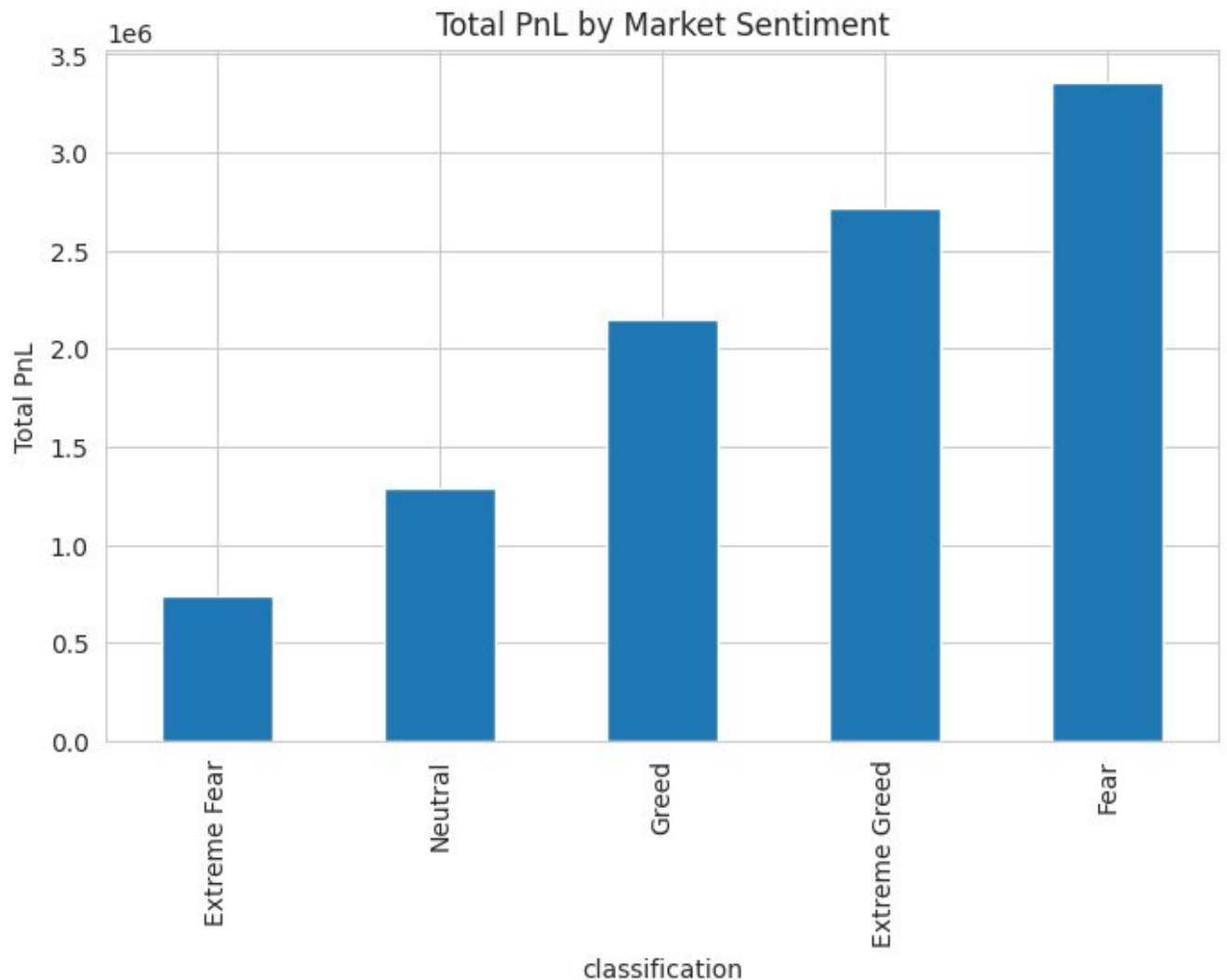
Finding 1: Extreme Greed Generates the Highest Average Profit

Extreme Greed achieved the highest average profit per trade (67.89). This suggests that strong bullish sentiment creates favorable trading opportunities and increased profitability for traders.



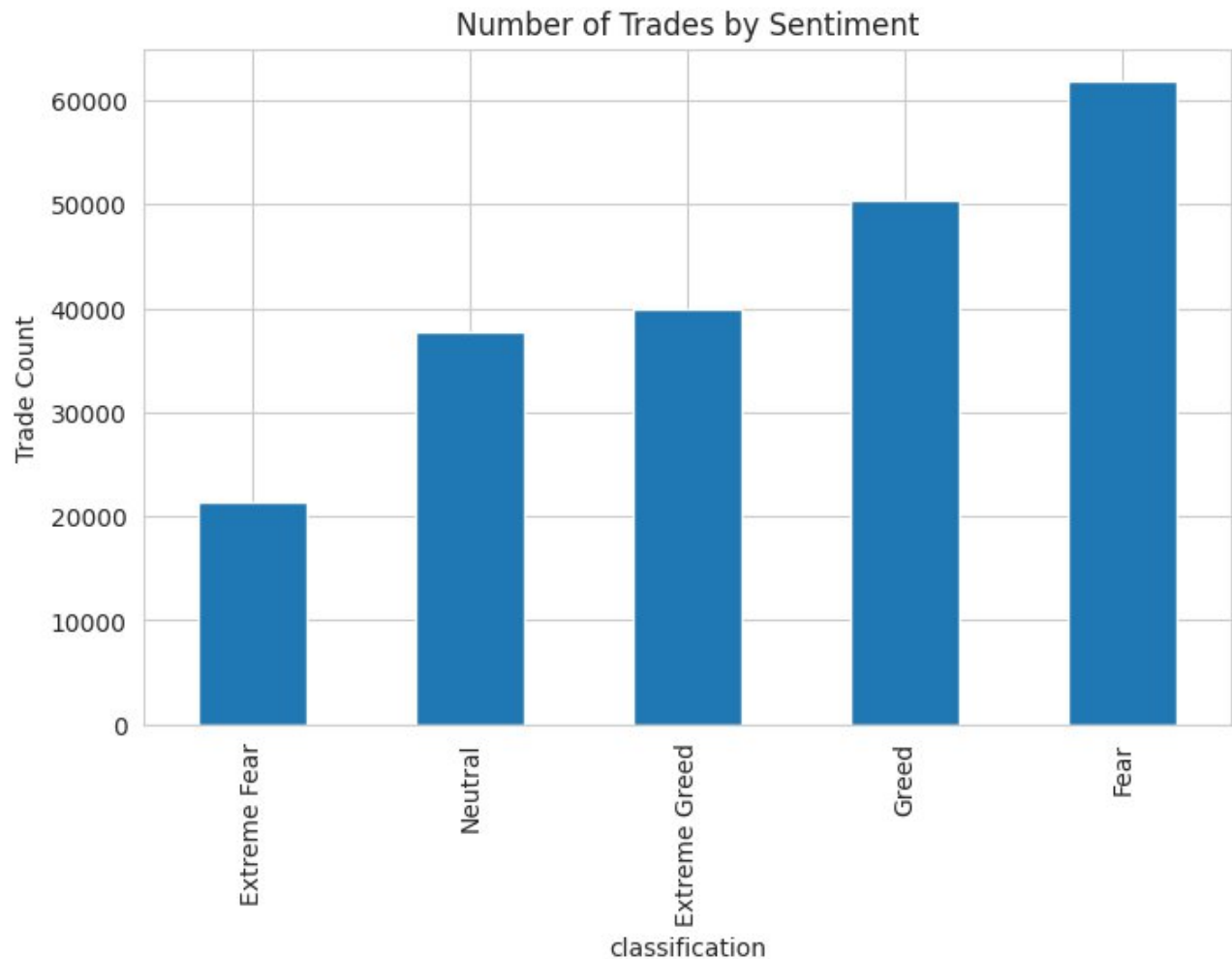
Finding 2: Fear Produces the Highest Total Profit

Fear generated the largest cumulative profit of approximately 3.36 million. This was also the sentiment category with the highest number of trades, indicating that traders become more active during fearful market conditions and successfully capitalize on increased volatility.



Finding 3: Trading Activity Peaks During Fear

The Fear category recorded over 61,000 trades, making it the most active market condition. Increased uncertainty appears to encourage more trading participation and opportunity-seeking behavior.

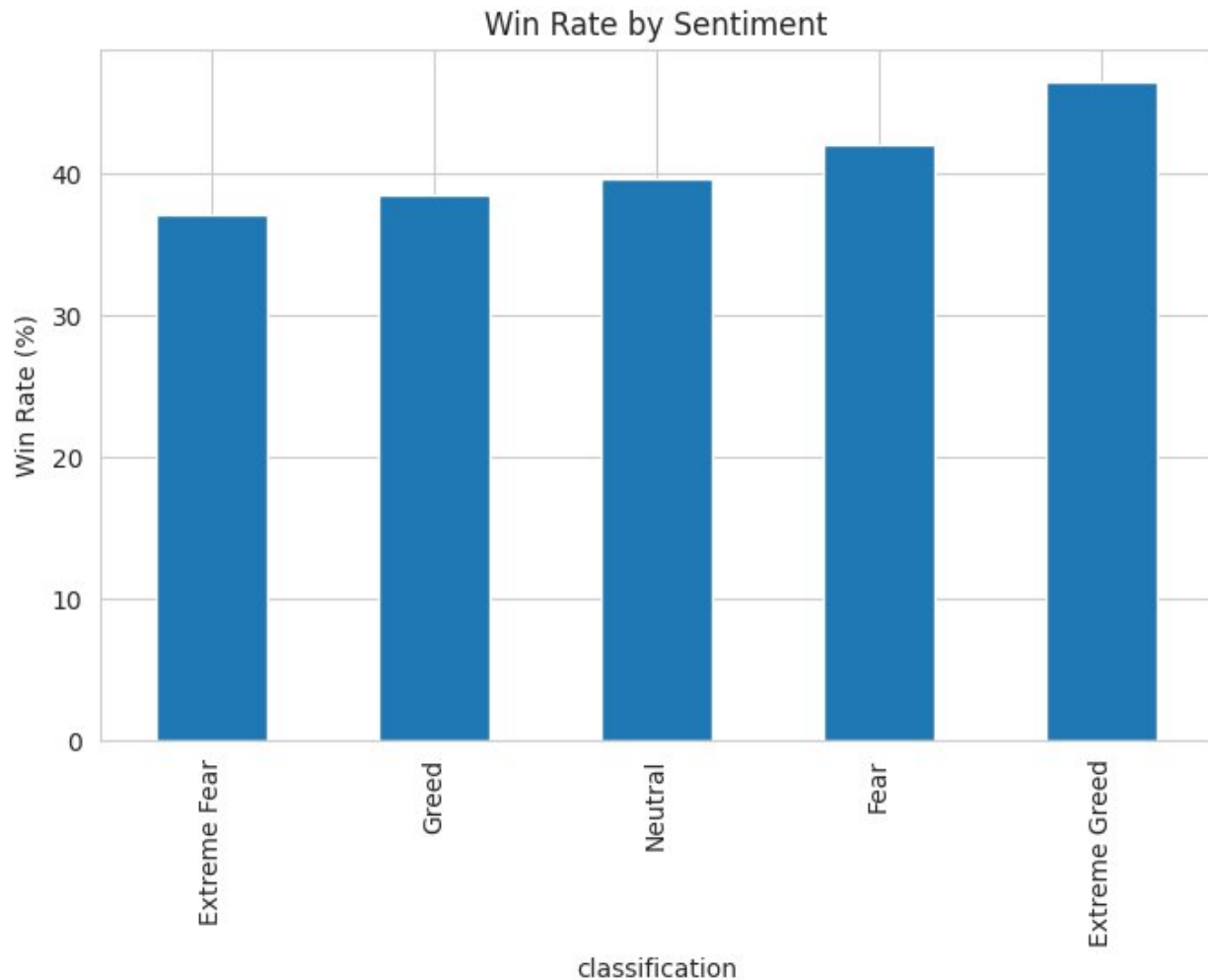


Finding 4: Neutral Markets Show Lower Profitability

Neutral market conditions resulted in one of the lowest average profits. This indicates that traders generally perform better when markets exhibit a strong directional bias rather than remaining indecisive.

Finding 5: Market Sentiment Influences Trader Performance

The analysis demonstrates a clear relationship between sentiment and profitability. Both Fear and Extreme Greed environments produced stronger trading outcomes than Neutral conditions.



5. Key Insights

1. Extreme Greed generated the highest average profit per trade.
2. Fear produced the highest total profit and trading activity.
3. Neutral markets showed comparatively weaker performance.
4. Traders appear to benefit from highly emotional market environments.
5. Market sentiment can be used as an additional factor in trading strategy design and decision-making.

6. Business Impact

The findings suggest that sentiment indicators can enhance trading strategies by:

- Identifying favorable market conditions.
- Improving trade timing decisions.
- Supporting risk management practices.
- Increasing awareness of market psychology.
- Enhancing profitability through sentiment-based decision making.

7. Conclusion

This study explored the relationship between Bitcoin market sentiment and trader performance using Hyperliquid trading data and the Fear & Greed Index. The results reveal that market sentiment significantly influences trader profitability and behavior.

Extreme Greed produced the highest average profit per trade, while Fear generated the largest overall profit due to increased trading activity. These findings highlight the importance of incorporating sentiment analysis into trading strategies. By combining sentiment indicators with traditional technical and quantitative methods, traders can make more informed decisions and potentially improve overall performance.

8. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

GitHub Repository

[bitcoin-sentiment-analysis](https://github.com/bitcoin-sentiment-analysis)